

PROYECTO 1

USO DE UN PROTOCOLO EXISTENTE

Análisis de la comunicación entre el servidor remoto y el cliente

Utilizando Wireshark se tiene el tráfico de red durante las interacciones de los servidores MCP, las cuales se observan a continuación. Se observa las interacciones del chatbot con los servidores de Google Cloud y Groq. Comienza, con la conexión inicial con la identificación de mensajes JSON-RPC (líneas 232, 357-362, 239-241, y entre otras). Luego, se observan mensajes de solicitud que se hacen a través de HTTP. Finalmente, los servidores devuelven la respuesta (también en un formato JSON-RP).

No.	Time	Source	Destination	Protocol	TCP Segment Len	Info
227	22:41:06.573489	fe80::22f3:75ff:feb...	ff02::1	ICMPv6		Router Advertisement from 20:f3:75:b3:74:50
228	22:41:07.056434	192.168.0.16	134.224.240.217	TLSv1.2	23	Application Data
229	22:41:08.212297	134.224.240.217	192.168.0.16	TCP	0	443 → 61192 [ACK] Seq=150 Ack=467 Win=9 Len=0
230	22:41:08.955122	192.168.0.16	134.224.240.217	TLSv1.2	210	Application Data
231	22:41:09.808506	134.224.240.217	192.168.0.16	TCP	0	443 → 61192 [ACK] Seq=150 Ack=677 Win=9 Len=0
232	22:41:09.808506	134.224.240.217	192.168.0.16	TLSv1.2	74	Application Data
233	22:41:09.808733	fe80::22f3:75ff:feb...	ff02::1	ICMPv6		Router Advertisement from 20:f3:75:b3:74:50
234	22:41:09.854102	192.168.0.16	134.224.240.217	TCP	0	61192 → 443 [ACK] Seq=677 Ack=224 Win=251 Len=0
235	22:41:12.096819	Comscope_b3:74:50	AzureKaveTec_bd:25:...	ARP		Who has 192.168.0.16? Tell 192.168.0.1
236	22:41:12.096866	AzureKaveTec_bd:25:...	Comscope_b3:74:50	ARP		192.168.0.16 is at 28:0b:43:bd:25:dd
237	22:41:12.577868	2803:d100:f390:6d6::...	2803:c800:0:78::2	DNS		Standard query 0x13a6 A unit-converter-mcp-304357449334.us-central1.run.app
238	22:41:12.924647	2803:c800:0:78::2	2803:d100:f390:6d6::...	DNS		Standard query response 0x13a6 A unit-converter-mcp-304357449334.us-central1.run.app CNAME v...
239	22:41:12.926409	2803:d100:f390:6d6::...	2600:1900:4245:200::...	TCP	0	54562 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM WS=256
240	22:41:13.429666	2600:1900:4245:200::...	2803:d100:f390:6d6::...	TCP	0	443 → 54562 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 SACK_PERM WS=256
241	22:41:13.429870	2803:d100:f390:6d6::...	2600:1900:4245:200::...	TCP	0	54562 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
242	22:41:13.822576	2600:1900:4245:200::...	2803:d100:f390:6d6::...	TCP	0	[TCP Out-Of-Order] 443 → 54562 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 SACK_PERM WS=...
243	22:41:13.822695	2803:d100:f390:6d6::...	2600:1900:4245:200::...	TCP	0	[TCP Dup. ACK] 54562 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0 SLE=0 SRE=1
244	22:41:14.268901	2803:d100:f390:6d6::...	2600:1900:4245:200::...	TLSv1.2	517	Client Hello (SHA1-unit-converter-mcp-304357449334.us-central1.run.app)
245	22:41:14.555892	2600:1900:4245:200::...	2803:d100:f390:6d6::...	TCP	0	443 → 54562 [ACK] Seq=1 Ack=518 Win=268800 Len=0
246	22:41:14.555892	2600:1900:4245:200::...	2803:d100:f390:6d6::...	TLSv1.3	1220	Server Hello, Change Cipher Spec
247	22:41:14.556358	2600:1900:4245:200::...	2803:d100:f390:6d6::...	TCP	1220	443 → 54562 [ACK] Seq=1221 Ack=518 Win=268800 Len=1220 [TCP PDU reassembled in 251]
248	22:41:14.556358	2600:1900:4245:200::...	2803:d100:f390:6d6::...	TCP	1220	443 → 54562 [ACK] Seq=2441 Ack=518 Win=268800 Len=1220 [TCP PDU reassembled in 251]
249	22:41:14.556358	2600:1900:4245:200::...	2803:d100:f390:6d6::...	TCP	1220	443 → 54562 [ACK] Seq=3661 Ack=518 Win=268800 Len=1220 [TCP PDU reassembled in 251]
250	22:41:14.556358	2600:1900:4245:200::...	2803:d100:f390:6d6::...	TCP	1220	443 → 54562 [PSH, ACK] Seq=4881 Ack=518 Win=268800 Len=1220 [TCP PDU reassembled in 251]
251	22:41:14.556358	2600:1900:4245:200::...	2803:d100:f390:6d6::...	TLSv1.3	700	Application Data
252	22:41:14.556454	2803:d100:f390:6d6::...	2600:1900:4245:200::...	TCP	0	54562 → 443 [ACK] Seq=518 Ack=6801 Win=65280 Len=0
253	22:41:14.559792	2803:d100:f390:6d6::...	2600:1900:4245:200::...	TLSv1.3	80	Change Cipher Spec, Application Data
254	22:41:14.560283	2803:d100:f390:6d6::...	2600:1900:4245:200::...	TLSv1.3	269	Application Data
255	22:41:14.560374	2803:d100:f390:6d6::...	2600:1900:4245:200::...	TLSv1.3	189	Application Data
256	22:41:15.438045	2803:d100:f390:6d6::...	2607:fb8b:400c:c0d::...	TCP	1	[TCP Keep-Alive] 50919 → 443 [ACK] Seq=1 Ack=1 Win=253 Len=1
257	22:41:15.442549	2600:1900:4245:200::...	2803:d100:f390:6d6::...	TCP	0	443 → 54562 [ACK] Seq=6801 Ack=1056 Win=268288 Len=0
258	22:41:15.442549	2600:1900:4245:200::...	2803:d100:f390:6d6::...	TLSv1.3	997	Application Data, Application Data
259	22:41:15.443571	2600:1900:4245:200::...	2803:d100:f390:6d6::...	TCP	997	[TCP Retransmission] 443 → 54562 [PSH, ACK] Seq=6801 Ack=1056 Win=268800 Len=997
260	22:41:15.443641	2803:d100:f390:6d6::...	2600:1900:4245:200::...	TCP	0	54562 → 443 [ACK] Seq=1056 Ack=7798 Win=64512 Len=0 SLE=6801 SRE=7798
261	22:41:15.444395	2803:d100:f390:6d6::...	2600:1900:4245:200::...	TCP	0	54562 → 443 [FIN, ACK] Seq=1056 Ack=7798 Win=64512 Len=0
262	22:41:15.831933	fe80::22f3:75ff:feb...	ff02::1	ICMPv6		Router Advertisement from 20:f3:75:b3:74:50
263	22:41:16.345821	2600:1900:4245:200::...	2803:d100:f390:6d6::...	TCP	0	443 → 54562 [FIN, ACK] Seq=7798 Ack=1057 Win=268800 Len=0
264	22:41:16.345963	2803:d100:f390:6d6::...	2600:1900:4245:200::...	TCP	0	54562 → 443 [ACK] Seq=1057 Ack=7799 Win=64512 Len=0
265	22:41:17.225509	192.168.0.1	224.0.0.1	IGMPv3		Membership Query, general
266	22:41:19.164731	fe80::22f3:75ff:feb...	ff02::1	ICMPv6		Router Advertisement from 20:f3:75:b3:74:50
267	22:41:19.478353	192.168.0.16	224.0.0.22	IGMPv3		Membership Report / Join group 224.0.0.252 for any sources / Join group 239.255.255.250 for ...

Figura 1. Wireshark comunicación 1

No.	Time	Source	Destination	Protocol	TCP Segment Len	Info
351	22:46:01.530898	18.97.36.19	192.168.0.16	TCP	0	443 → 53834 [ACK] Seq=143 Ack=653 Win=414 Len=0
352	22:46:01.530898	18.97.36.19	192.168.0.16	TLSv1.2	118	Application Data
353	22:46:01.578010	192.168.0.16	18.97.36.19	TCP	0	53834 → 443 [ACK] Seq=653 Ack=261 Win=255 Len=0
354	22:46:03.724469	fe80::22f3:75ff:feb...	ff02::1	ICMPv6		Router Advertisement from 20:f3:75:b3:74:50
355	22:46:03.747104	2803:d100:f390:6d6::...	2606:4700:4400::681...	TCP	0	53801 → 443 [FIN, ACK] Seq=1331 Ack=5657 Win=65280 Len=0
356	22:46:03.748671	2803:d100:f390:6d6::...	2803:c800:0:78::12	DNS		Standard query 0xbf89 AAAA api.groq.com
357	22:46:03.749086	2803:d100:f390:6d6::...	2803:c800:0:78::12	DNS		Standard query 0xbf89 AAAA api.groq.com
358	22:46:04.069798	192.168.0.16	10.240.80.254	DNS		Standard query 0xbf89 AAAA api.groq.com
359	22:46:04.069906	192.168.0.16	10.240.80.254	DNS		Standard query 0x5953 A api.groq.com
360	22:46:04.071031	2606:4700:4400::681...	2803:d100:f390:6d6::...	TCP	0	443 → 53801 [FIN, ACK] Seq=5656 Ack=1331 Win=131072 Len=0
361	22:46:04.071031	2803:c800:0:78::12	2803:d100:f390:6d6::...	DNS		Standard query response 0xbf89 AAAA api.groq.com AAAA 2606:4700:4400::6812:2862 AAAA 2606:47...
362	22:46:04.071031	2803:c800:0:78::12	2803:d100:f390:6d6::...	DNS		Standard query response 0x5953 A api.groq.com A 172.64.147.158 A 104.18.40.98
363	22:46:04.071239	2803:d100:f390:6d6::...	2606:4700:4400::681...	TCP	0	53801 → 443 [ACK] Seq=1331 Ack=5657 Win=65280 Len=0
364	22:46:04.072536	2803:d100:f390:6d6::...	2606:4700:4400::681...	TCP	0	57383 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
365	22:46:04.481037	2606:4700:4400::681...	2803:d100:f390:6d6::...	TCP	0	443 → 57383 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1360 SACK_PERM WS=8192
366	22:46:04.481037	10.240.80.254	192.168.0.16	DNS		Standard query response 0xbf89 AAAA api.groq.com AAAA 2606:4700:4400::6812:2862 AAAA 2606:47...
367	22:46:04.481037	10.240.80.254	192.168.0.16	DNS		Standard query response 0x5953 A api.groq.com A 172.64.147.158 A 104.18.40.98
368	22:46:04.481239	2803:d100:f390:6d6::...	2606:4700:4400::681...	TCP	0	57383 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
369	22:46:04.504541	2803:d100:f390:6d6::...	2606:4700:4400::681...	TLSv1.2	517	Client Hello (SHI=api.groq.com)
370	22:46:04.888412	2606:4700:4400::681...	2803:d100:f390:6d6::...	TCP	0	443 → 57383 [ACK] Seq=1 Ack=518 Win=131072 Len=0
371	22:46:04.888412	2606:4700:4400::681...	2803:d100:f390:6d6::...	TLSv1.3	2720	Server Hello, Change Cipher Spec
372	22:46:04.888412	2606:4700:4400::681...	2803:d100:f390:6d6::...	TLSv1.3	108	Application Data
373	22:46:04.888570	2803:d100:f390:6d6::...	2606:4700:4400::681...	TCP	0	57383 → 443 [ACK] Seq=518 Ack=2829 Win=65280 Len=0
374	22:46:04.894131	2803:d100:f390:6d6::...	2606:4700:4400::681...	TLSv1.3	80	Change Cipher Spec, Application Data
375	22:46:04.895236	2803:d100:f390:6d6::...	2606:4700:4400::681...	TLSv1.3	793	Application Data
376	22:46:04.895601	2803:d100:f390:6d6::...	2606:4700:4400::681...	TCP	1360	57383 → 443 [ACK] Seq=1391 Ack=2829 Win=65280 Len=1360 [TCP PDU reassembled in 377]
377	22:46:04.895601	2803:d100:f390:6d6::...	2606:4700:4400::681...	TLSv1.3	884	Application Data
378	22:46:04.968808	2606:4700:4400::681...	2803:d100:f390:6d6::...	TCP	0	443 → 57383 [ACK] Seq=2829 Ack=2751 Win=139264 Len=0
379	22:46:05.499977	192.168.0.16	18.97.36.5	TLSv1.2	240	Application Data
380	22:46:05.500914	2606:4700:4400::681...	2803:d100:f390:6d6::...	TCP	0	443 → 57383 [ACK] Seq=2829 Ack=3635 Win=139264 Len=0
381	22:46:06.014618	18.97.36.5	192.168.0.16	TCP	0	443 → 64055 [ACK] Seq=119 Ack=626 Win=284 Len=0
382	22:46:06.014618	18.97.36.5	192.168.0.16	TLSv1.2	118	Application Data
383	22:46:06.060645	192.168.0.16	18.97.36.5	TCP	0	64055 → 443 [ACK] Seq=626 Ack=237 Win=252 Len=0
384	22:46:06.060958	2606:4700:4400::681...	2803:d100:f390:6d6::...	TLSv1.3	1360	Application Data
385	22:46:06.060958	2606:4700:4400::681...	2803:d100:f390:6d6::...	TLSv1.3	529	Application Data
386	22:46:06.060949	2803:d100:f390:6d6::...	2606:4700:4400::681...	TCP	0	57383 → 443 [ACK] Seq=3635 Ack=4718 Win=65280 Len=0
387	22:46:06.060958	2606:4700:4400::681...	2803:d100:f390:6d6::...	TLSv1.3	656	Application Data
388	22:46:06.060958	2606:4700:4400::681...	2803:d100:f390:6d6::...	TLSv1.3	37	Application Data
389	22:46:06.060958	2606:4700:4400::681...	2803:d100:f390:6d6::...	TLSv1.3	27	Application Data
390	22:46:06.060958	2803:d100:f390:6d6::...	2606:4700:4400::681...	TCP	0	57383 → 443 [ACK] Seq=3635 Ack=5438 Win=64768 Len=0
391	22:46:06.921198	fe80::22f3:75ff:feb...	ff02::1	ICMPv6		Router Advertisement from 20:f3:75:b3:74:50

Figura 1. Wireshark comunicación 2

MCP desarrollados

- MCP películas: su endpoint es un local host en el puerto 8000. Además, tiene un motor de búsqueda para detallar información de la película, también recomienda películas y da películas aleatorias. Sus parámetros son: título de la película, géneros de películas y rating.
- MCP trivial: su endpoint es por medio de Google Cloud. Sus implementaciones más importantes son conversiones de unidades de medida. Sus parámetros son los valores a cambiar, la unidad de origen y la unidad deseada.

Explicación a nivel de capas

Para las capas de red de la comunicación del MCP sucede lo siguiente:

- Capa de enlace: aquí se usa el ethernet para la comunicación local entre las direcciones.
- Capa de red: usa los protocolos (IPv4 e IPv6) para el enrutamiento de paquetes través de Internet hacia los servidores remotos.
- Capa de transporte: usa el protocolo TCP para asegurar una conexión estable y confiable utilizando puertos dinámicos.
- Capa de aplicación: se comunica en base a JSON-RPC y HTTP cifrado con TLS 1.3.

Conclusiones y comentarios

- Por medio de este proyecto pude entender la importancia de una buena estructura y arquitectura de los códigos y requerimientos. Ya que, sin ello no era posible tener un MCP confiable, debido a que es importante entre la comunicación de la red.

- Fue difícil entender los pasos para realizar un MCP porque se debía de instalar varias dependencias.